

Network Intrusion Detection System Based on Enhanced Dual Gaussian Mixture Variational Autoencoders for Internet of Vehicles

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Abstract—The Internet of Vehicles (IoV) is highly vulnerable to attacks due to its open communication environment, with new types of attacks continuously emerging. However, existing Network Intrusion Detection Systems (NIDS) often fall short in accuracy, recall rates, false positive rates, and they frequently prove ineffective in detecting new attacks. In this paper, we propose a NIDS for IoV based on Enhanced Dual Gaussian Mixture Variational Autoencoders (EDGMVAE) to detect various attack behaviors, including new attacks. We employ two separate Gaussian Mixture Variational Autoencoders (GMVAE) to conduct unsupervised reconstruction training on normal traffic and known attack traffic. During detection, we obtain the tested traffic's reconstruction probabilities using these GMVAEs and determine whether an attack has occurred through logical fusion operation. We use two datasets for evaluation, i.e., the Car Hacking Dataset (CHD) for in-vehicle communication and the UNSW-NB15 Dataset (UBD) for external network communication. The results demonstrate that our method achieves remarkable performance in detecting known attacks, with an accuracy rate of 98.54% on the CHD and 98.59% on the UBD. Moreover, it outperforms other state-of-the-art methods in detecting new attacks. Specifically, on the CHD, the accuracy of the method is 6.66% to 12.83% higher than other methods, while on the UBD, the accuracy is 5.13% to 21.91% higher than other methods.

Index Terms—Internet of vehicles, intrusion detection, new attack, Gaussian mixture variational autoencoder.

I. INTRODUCTION

INTERNET of Vehicles (IoV) is an integral component of Intelligent Transportation System (ITS). Leveraging advanced information and communication technologies, IoV facilitates comprehensive network connectivity within vehicles, as well as Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P), and Vehicle-to-Service (V2S) communication [1]. This connectivity not only enhances the overall intelligent driving capabilities of vehicles but also delivers a safe, comfortable, intelligent, and efficient driving experience and traffic services to users [2].

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However, the open communication environment inherent to IoV also renders it a prime target for attackers, which can lead to severe repercussions, as malicious attacks on vehicles traveling on the road would pose a serious threat to human life. Threats to IoV are divided into two main categories: (1) Attacks on vehicle-to-external networks; (2) Attacks on in-vehicle networks.

When a vehicle connects to external networks, it becomes vulnerable to various attacks, including Denial of Service (DoS), Distributed Denial of Service (DDoS), Spoofing and Replay. In DoS and DDoS attacks, malicious actors overwhelm vehicle nodes by sending a massive volume of irrelevant messages. This deluge of traffic congests the vehicle's processing capacity, rendering it incapable of handling legitimate requests. Attackers can also impersonate legitimate users to transmit fake GPS data to the vehicle, with the intention of deceiving it. Replay attacks intercept communication to identify unencrypted data, enabling hackers to conduct Man-in-the-Middle (MITM) attacks.

In addition to external attacks, vehicles are also at risk from internal attacks. The Controller Area Network (CAN) is the standard communication method for in-vehicle networks, connecting various Electronic Control Units (ECUs) to support critical applications such as autonomous driving. However, the lack of authentication mechanisms makes it easy for attackers to launch attacks. Common types of attacks include Fuzzy, DoS and spoofing. Fuzzy attacks cause malfunctions or abnormalities in the vehicle system by sending a large amount of random information. DoS attacks overload the target system, blocking normal services and rendering some of its functions unusable. Spoofing attacks can deceive vehicle systems by transmitting false gear or speed data, forcing incorrect decisions and potentially triggering safety incidents. With the rapid development of technology, the attack methods IoV faces are also constantly evolving [3]. In order to protect the IoV from attacks from both inside and outside, it is necessary to develop a powerful intrusion detection system that can not only accurately detect known attack patterns but also effectively handle emerging attacks.

Currently, Machine Learning (ML) and Deep Learning (DL) methods have become pivotal technologies in the development of intrusion detection systems due to their advanced learning capabilities [4], [5], [6], [7]. These methods generally necessitate supervised training of binary or multi-classification

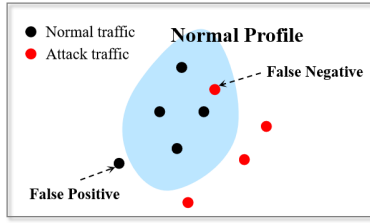


Fig. 1. Challenges faced by ANIDS.

models, utilizing extensive datasets that encompass both normal and attack traffic, in order to attain high detection performance. However, most existing ML and DL methods prove ineffective in detecting emerging new attacks due to the absence of representative features of these attacks in the training datasets. Thus, Anomaly-based NIDS (ANIDS) is the most suitable option, which can detect existing and new attacks as it builds a profile from normal data and any deviation from that profile is detected as an anomaly [8], [9]. However, ANIDS also confronts several significant challenges, as illustrated in Fig. 1. Firstly, it is difficult for normal profile to fully cover all normal traffic patterns, which will inevitably lead to a high false positive rate. In addition, as the technical sophistication of attackers increases, their attack behavior will become more similar to normal traffic patterns, making it harder to detect and leading to inevitable false negatives.

To address the above challenges, we propose an ANIDS based on Enhanced Dual Gaussian Mixture Variational Autoencoders (EDGMVAE) for IoV. The high-level design goals of our method are twofold: (1) Our primary goal is to model normal behavior patterns with the utmost precision and comprehensiveness, which is essential for effectively reducing false positives. (2) We add the profiling of known attacks as a reference in the detection process to reduce false negatives, especially for attacks whose behavioral patterns closely resemble normal traffic. When building traffic profiles, we propose using Gaussian Mixture Variational Autoencoder (GMVAE), which is an improvement upon the traditional Variational Autoencoder (VAE). Specifically, the GMVAE replaces the single Gaussian distribution of latent variables with a mixture of Gaussians, thereby achieving more accurate and comprehensive modeling capabilities. This improvement allows for arbitrary precision in approximating any distribution by simply adding or removing components (i.e., the number of Gaussian distributions).

Our EDGMVAE approach comprises three phases. In the first phase, raw traffic data is processed to extract and standardize key features, thereby ensuring data quality. In the second phase, the processed data is fed into two GMVAE models: the GMVAE_N (trained on normal traffic to build normal profile), and GMVAE_A (trained on known attack traffic to build attack profile), both of which are used for data reconstruction. In the third phase, the reconstruction probabilities of the input data are computed using the two GMVAEs separately, and the final detection results are obtained through logical fusion operations. We comprehensively evaluate the proposed EDGMVAE on two datasets: the Car Hacking Dataset (CHD) [10] for in-vehicle communication and the UNSWNB-15 Dataset

(UBD) [11] for external network communication. The results demonstrate that our method achieves high accuracy in detecting known attacks: 98.54% on the CHD and 98.59% on the UBD. It also outperforms other state-of-the-art methods in detecting new attacks, with accuracy improvements of 6.66%-13.61% on the CHD and 5.13%-21.91% on the UBD. These results further confirm the excellent performance and advantages of EDGMVAE in detecting new attacks. The primary contributions of this work are summarized as follows:

- We propose a network intrusion detection system for IoV based on EDGMVAE. This method aims to accurately detect both known and emerging new attack patterns by utilizing the strengths of GMVAE.
- We implement the EDGMVAE approach with a three-phase framework, including data preprocessing, data reconstruction, and detection.
- We conduct several experiments on the CHD and UBD to demonstrate that our EDGMVAE approach not only achieves excellent performance in detecting known attacks but also outperforms other compelling methods in detecting new attacks.

The article is organized as follows: in Section I, we provide an introduction, and in Section II, we examine relevant literature. Section III provides research motivations and model preliminaries. Our proposed approach is presented in Section IV. We present experimental verification and assessment in Section V. Finally, in Section VI, we conclude the paper.

II. RELATED WORK

With the rapid development of the IoV, ensuring network security has become increasingly critical, prompting extensive research into IDS. To systematically summarize existing efforts, Table I provides a comparative overview of representative studies, evaluating them based on detection scope (in-vehicle vs. external networks), attack types (known vs. unknown), datasets and technical mechanisms.

A. In-Vehicle Network Intrusion Detection Systems

A significant amount of research has been dedicated to securing in-vehicle networks, with a particular focus on the CAN bus. Early efforts in this domain primarily targeted known attacks, leveraging classical deep learning models to achieve high detection accuracy. Hossain et al. [12] pioneered this direction by proposing an LSTM-based NIDS for CAN bus attacks. They constructed a custom dataset by extracting normal data from a Toyota hybrid vehicle and injecting DoS, Fuzzing, and Spoofing attacks, achieving an impressive 99.995% accuracy in detecting these known threats.

Wei et al. [13] introduced AMAEID, an intrusion detection model that combines an attention mechanism with a multi-layer denoising autoencoder. Xie et al. [14] proposed an intrusion detection technique based on an enhanced GAN. It aims to solve the problems of existing GAN models, such as the rough construction of CAN message blocks and the inability to detect data tampering threats, thereby improving the detection performance against various attacks. Song et al. [17] proposed an IDS based on a deep convolutional neural

TABLE I
COMPARATIVE EVALUATION OF THE EXISTING RESEARCH RELATED TO IDS FOR IOV

Paper	In-vehicle network intrusion detection	External network intrusion detection	New attacks detection	Computational overhead analysis	Detection mechanism
Hossain et al. [12]	✓				LSTM
Wei et al. [13]	✓				Attention & Autoencoder
Xie et al. [14]	✓				GAN
luo et al. [15]	✓		✓		SR-LiCNN
Shi et al. [16]	✓		✓		Deep embedded clustering
Song et al. [17]	✓				DCNN
Jin et al. [18]	✓	✓			Outlier detection
Ullah et al. [19]	✓	✓			Ensemble approach
Song et al. [20]	✓		✓		LSTM & ResNet
Cao et al. [21]	✓		✓	✓	MPMHit
Yang et al. [22]	✓	✓	✓		Hybrid approach
Almutlaq et al. [23]	✓	✓		✓	Rule-based
Huang et al. [24]		✓	✓	✓	Ensemble approach
Khan et al. [25]	✓	✓	✓		Bloom Filter & Bi-LSTM
Anbalagan et al. [26]		✓		✓	CNN
Nissar et al. [27]		✓	✓		VAE
Fu et al. [28]	✓	✓	✓		BERT
This paper	✓	✓	✓	✓	EDGMVAE

network (DCNN). Through a frame builder, the 29-bit IDs of 29 consecutive CAN messages are converted into a 29×29 two-dimensional grid structure, which serves as input to a reduced Inception-ResNet model to learn the temporal patterns of CAN traffic and detect anomalies. Jin et al. [18] proposed an intrusion detection approach that integrates oversampling, outlier detection, and metric learning techniques. Ullah et al. [19] proposed a stacking-based hybrid ensemble model employing XGBoost, Random Forest, Decision Tree, and Gradient Boosting as base learners, integrated with K-means sampling, Information Gain feature selection, and SMOTE to address class imbalance, achieving high performance in Internet of Vehicles intrusion detection.

Recent advancements have expanded capabilities to include unknown attack detection, driven by innovations in model architecture and data processing. Song et al. [20] adopted a self-supervised approach, combining an LSTM generator (trained on normal traffic to produce noisy pseudo-normal data) with a Reduced Inception-ResNet detector, allowing the model to generalize to unseen threats. Luo et al. [15] proposed a spectral residual-lightweight CNN (SR-LiCNN) model, which transforms signal sequences into matrix-like images to frame intrusion detection as an image classification task. The spectral residual operation eliminates redundancy, while lightweight convolutions optimize decision-making efficiency, enabling detection of both known and unknown attacks.

Shi et al. [16] proposed an unsupervised intrusion detection system IDS-DEC for the CAN bus, which is based on deep embedded clustering and integrates a spatiotemporal autoencoder with an entropy-regularized deep embedding clustering module. Cao et al. [21] proposed a self-supervised

learning-based anomaly detection method for in-vehicle networks, which enables message-level anomaly detection without labeled data through a message prediction model based on hierarchical Transformers (MPMHit). The method consists of three modules: masking, predicting, and anomaly detecting. It uses noisy message sequences to train the model to learn the distribution of normal messages and combines an online update mechanism with vehicle-cloud collaboration to address the problem of concept drift.

B. Vehicle-to-External Network Intrusion Detection Systems

As the IoV becomes increasingly interconnected, vehicle-to-external network communications have become a key security focus. The complexity and variety of potential attacks on these networks promotes the continuous development of intrusion detection technologies.

Early research on vehicle-to-external network intrusion detection mainly focused on known attacks. Almutlaq et al. [23] proposed a two-stage intrusion detection system that balances detection accuracy, resource efficiency, and interpretability by extracting rules from deep neural networks. The first stage of the system distinguishes between normal and attack traffic, while the second stage identifies the type of attack. It includes three variants based on the DeepRed and HypInv rule extraction methods. Anbalagan et al. [26] proposed a CNN-based NIDS. First, the original data captured was processed, and quantile transformation was applied to obtain uniformly distributed labels. Then, the processed data was converted into image data for fine-grained pattern extraction. Subsequently, an improved deep CNN architecture

was used to train the converted image data to identify various attack patterns.

To overcome the limitations of only detecting known attacks, researchers have conducted extensive exploration and developed systems that can detect both known and new attacks by adopting more advanced technologies. Khan et al. [25] proposed a multi-stage deep learning framework. This framework's first stage verified the status of incoming data using state-based Bloom filter technology, and its second stage detected threats using a Bi-LSTM classifier. Nissar et al. [27] proposed a multi-objective intrusion detection framework that incorporated Variational Autoencoder (VAE) along with two optimization algorithms: the Adaptive Geometric Estimation Multi-Objective Evolutionary Algorithm (AGE-MOEA) and the Reference Point-based NSGA-III (R-NSGA-III). This framework was designed to optimize multiple objectives, including divergence, Kullback-Leibler (KL) divergence, and reconstruction error, effectively detecting anomalies based on reconstruction error.

Yang et al. [22] proposed a multi-tiered hybrid intrusion detection system (MTH-IDS) for the Internet of Vehicles (IoV), aiming to detect both known and unknown attacks. The system includes data preprocessing, feature engineering, and a four-tier learning model. It detects known attacks through tree-based algorithms, identifies unknown attacks using cluster labeling k-means, and combines methods such as Bayesian optimization to improve performance. Huang et al. [24] proposed an Automatic Incremental Lifetime Learning Intrusion Detection System (AILL-IDS), which reduces the reliance on labeled data through incremental semi-supervised learning and triggers model updates by detecting concept drift with the KSWIN algorithm. The system combines incremental k-means, DBSTREAM clustering algorithms, and a dual ensemble learning mechanism to achieve anomaly detection, attack type identification, and continuous detection during updates, avoiding performance fluctuations. Fu et al. [28] proposed a BERT-based hybrid network intrusion detection system for the Internet of Vehicles (IoV-BERT-IDS), which converts traffic data into contextual byte sentences through a semantic extractor and combines pre-training (including MBWM and NBSP tasks) with a fine-tuning mechanism to achieve attack detection.

III. MOTIVATIONS AND PRELIMINARIES

A. Motivations

Through analysis of existing Anomaly-based NIDS (ANIDS), we find that they have advantages in detecting new attacks. However, ANIDS suffer from high false positive and false negative rates. This is mainly because constructing a precise and comprehensive normal profile is challenging, which can lead to false positives, and some attack patterns are extremely similar to normal patterns, which can lead to false negatives. If we can construct the normal profile as accurately and comprehensively as possible, and incorporate the profile of known attack patterns into the detection process, it will significantly improve the detection performance. When building profiles, we introduce the Gaussian Mixture Variational

Autoencoder (GMVAE), an enhanced variant of the Variational Autoencoder (VAE). By replacing VAE's single Gaussian latent variable distribution with a mixture of Gaussians, we can achieve more flexible and accurate performance of profile construction. Based on this improvement, we propose a NIDS leveraging Enhanced Dual Gaussian Mixture Variational Autoencoder (EDGMVAE), aiming to significantly enhance intrusion detection ability.

B. Model Preliminaries

1) *Variational Autoencoder (VAE)*: VAE is a deep generative model widely used for data compression, denoising, anomaly detection and generation. The core concept of a VAE involves mapping input data x to a low-dimensional latent space z via an encoder and then reconstructing the input data from this space using a decoder. This process enables effective data representation and facilitates the generation of new data samples similar to the input data.

VAE is understood through two perspectives: variational inference and neural network. The core objective is to obtain the posterior distribution $P(z|x)$, with x as the input data and z as the latent variable, and $P(z|x)$ can be calculated using Bayes's theorem:

$$P(z|x) = \frac{P(x|z)P(z)}{P(x)}. \quad (1)$$

However, direct computation of the marginal probability $P(x) = \int P(x|z)P(z)dz$ is generally intractable. Thus, variational inference addresses this problem by introducing a tractable distribution $Q_\phi(z|x)$ (often Gaussian) to approximate $P(z|x)$. The Kullback-Leibler (KL) divergence between these distributions must be minimized, as shown below:

$$D_{KL}[Q_\phi(z|x)||P(z|x)] = \mathbb{E}_{Q_\phi(z|x)}[\log Q_\phi(z|x) - \log P(z|x) - \log P(z) + \log P(x)]. \quad (2)$$

$\log P(x)$ can be shifted to the left since the expectation is with regard to z :

$$\begin{aligned} D_{KL}[Q_\phi(z|x)||P(z|x)] - \log P(x) \\ = - \underbrace{\mathbb{E}_{Q_\phi(z|x)}[\log P(x, z) - \log Q_\phi(z|x)]}_{\text{ELBO}(\phi)}. \end{aligned} \quad (3)$$

The objective of minimizing the KL divergence is equivalent to maximizing the Evidence Lower Bound (ELBO). The expression in (3) can be reformulated as:

$$\begin{aligned} D_{KL}[Q_\phi(z|x)||P(z|x)] - \log P(x) \\ = D_{KL}[Q_\phi(z|x)||P(z)] - \mathbb{E}_{Q_\phi(z|x)}[\log P(x|z)]. \end{aligned} \quad (4)$$

Then the role of neural network becomes evident, as depicted in Fig. 2a. $Q_\phi(z|x)$ serves as an encoder, converting the original data x into the latent variable z . Meanwhile, $P_\theta(x|z)$ functions as a decoder, which decodes z to generate reconstructed data x' . The objective of the model is to minimize the right part of (4). KL divergence between $Q_\phi(z|x)$ and $P(z)$ is represented by the first component on the right. The following component measures the degree of similarity between the original and the reconstructed data.

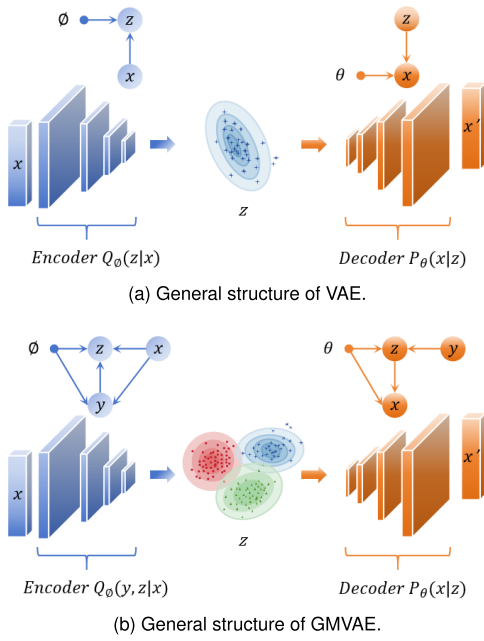


Fig. 2. Structure of VAE and GMVAE.

2) *Gaussian Mixture Variational Autoencoder (GMVAE)*: VAEs encounter challenges when processing complex data. This is due to the fact that the approximate posterior distribution and the prior distribution of the latent variable z are typically simple Gaussian distributions. These simple distributions often fail to adequately capture the complex characteristics of the original data, causing the reconstructed data to be both unrealistic and incomplete. It is theoretically possible to represent any complex data distribution using Gaussian Mixture distribution. Consequently, incorporating Gaussian Mixtures into VAE to construct Gaussian Mixture Variational Autoencoder (GMVAE) can significantly enhance the model's ability to represent and reconstruct complex data.

Specifically, we set the estimated posterior distribution $Q_\phi(y, z|x)$ and the posterior distribution $P_\theta(z)$ of the latent variable z as Gaussian Mixture distribution, where the Gaussian distribution category is represented by the variable y . The expression for $Q_\phi(y, z|x)$ is as follows:

$$Q_\phi(y, z|x) = Q_\phi(y|x)Q_\phi(z|x, y). \quad (5)$$

A Gaussian Mixture with k distinct Gaussian distributions makes up the latent variable. It is assumed that $Q_\phi(z|x, y_i)$ is a single Gaussian distribution with $i \in \{0, 1, \dots, k-1\}$. It is worth noting that an appropriate value of k needs to be selected. A too small k will result in the Gaussian components being unable to cover the complex patterns of the data, while a too large k will increase model redundancy, which not only raises computational costs but also may lead to overfitting. ELBO is reformulated as:

$$ELBO = \mathbb{E}_{Q_\phi(y, z|x)} [\log P_\theta(x, y, z) - \log Q_\phi(y, z|x)]. \quad (6)$$

According to conditional independence hypothesis shown in Fig. 2b, $P_\theta(x, y, z)$ is:

$$P_\theta(x, y, z) = P_\theta(x|z)P_\theta(z|y)P_\theta(y). \quad (7)$$

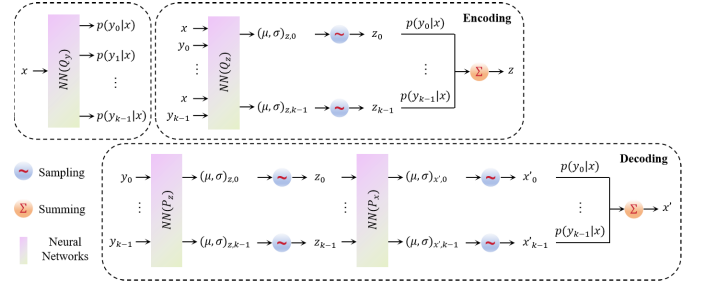


Fig. 3. Process of GMVAE.

TABLE II
DISTRIBUTIONS IN GMVAE MODEL

Encoder	Decoder
$Q(y x) = Multinomial(f(x))$	$P(y) = Uniform(\frac{1}{k})$
$Q(z x, y) = \mathcal{N}(\mu_z(x, y), \sigma_z^2(x, y))$	$P(z y) = \mathcal{N}(\mu_z(y), \sigma_z^2(y))$
	$P(x z) = \mathcal{N}(\mu_x(z), \sigma_x^2(z))$

Equation (5) and (7) can be substituted into (6) to reformulate ELBO as:

$$ELBO = \mathbb{E}_{Q_\phi(y, z|x)} \left[\log P(y) - \log Q_\phi(y|x) + \log \frac{P_\theta(z|y)}{Q_\phi(z|x, y)} + \log P_\theta(x|z) \right] \quad (8)$$

The third and fourth components in equation (8) represent the regularization and reconstruction, respectively. $\mathbb{E}_{Q_\phi(y, z|x)} [\log Q_\phi(y|x)]$ is a conditional entropy, capturing the degree to which x influences y . During training, a coefficient α can be added to the entropy term between x and y to regulate the influence of clustering loss items on other loss items:

$$ELBO = \mathbb{E}_{Q_\phi(y, z|x)} \left[\log P(y) - \alpha \log Q_\phi(y|x) + \log \frac{P_\theta(z|y)}{Q_\phi(z|x, y)} + \log P_\theta(x|z) \right]. \quad (9)$$

A smaller α reduces the impact of the entropy term on the loss, resulting in less distinct boundaries between Gaussian components. Conversely, a larger α strengthens this influence, leading to sharper boundaries, but also increasing the risk of overfitting.

Table II lists the probability distributions used in the model, while the detailed process of GMVAE is depicted in Fig. 3. Initially, x is assigned into k clusters based on the probabilities provided by $NN(Q_y)$, while $Q(y|x)$ denotes the probabilities of these cluster assignments and follows a multinomial distribution. $NN(Q_z)$ is used to learn the variance σ and mean μ of these Gaussians, since each cluster's latent space is thought to follow a Gaussian distribution. Then, by collecting samples and calculating their anticipated values, z is derived as $z = \sum_{i=0}^{k-1} p(y_i|x)z_i$.

During the decoding process, $NN(P_z)$ extracts the variance and mean from the relevant category. After that, reparameterization sampling is used to obtain the latent variable z , where $P(y)$ has a uniform distribution. The reconstructed data x' is

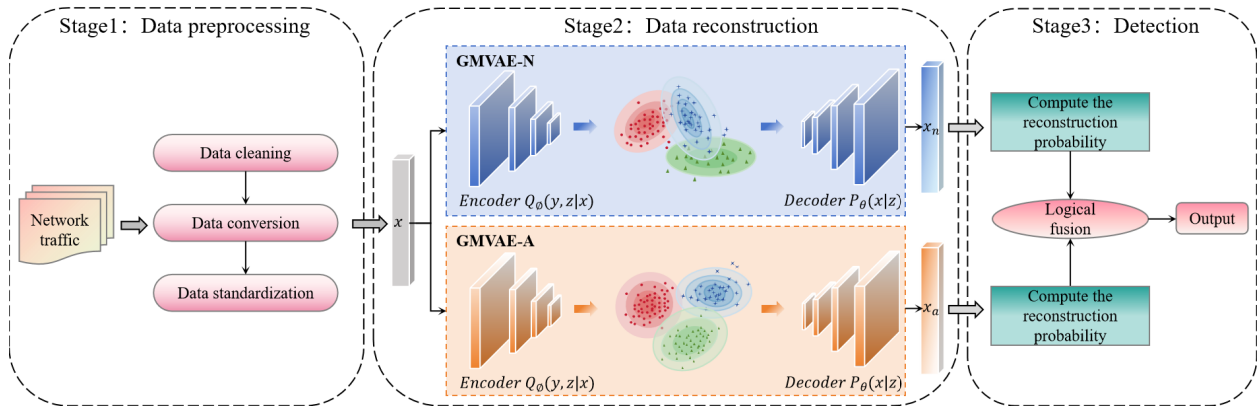


Fig. 4. Workflow of proposed EDGMVAE.

then calculated using z_i by sampling and taking the expected value of x'_i , which is $x' = \sum_{i=0}^{k-1} P(y_i|x)x'_i$.

IV. PROPOSED METHOD

A. Overview

GMVAE enhances the modeling capabilities of traditional VAE for complex data by introducing a Gaussian mixture latent distribution. However, employing a single GMVAE in NIDS scenarios presents notable shortcomings. Specifically, when using a single GMVAE, the profile is constructed exclusively for normal traffic. When attack traffic closely resembles normal patterns, the single model finds it difficult to detect subtle differences due to the absence of a reference attack profile, resulting in a high risk of false negatives. To address this issue, we propose a novel framework called Enhanced Dual Gaussian Mixture Variational Autoencoder (EDGMVAE). This framework builds independent profiles for normal traffic and known attack traffic using two GMVAEs. By combining their reconstruction probabilities, it enables more accurate detection performance.

The proposed EDGMVAE is depicted in Fig. 4 and comprises three primary stages, i.e., data preprocessing, data reconstruction, and detection. First, we preprocess the network traffic data. Subsequently, the preprocessed data x is simultaneously fed into two GMVAEs for data reconstruction. Specifically, GMVAE_N is a model that has been trained on normal traffic data, and GMVAE_A has been trained on known attack traffic data. During the detection stage, the reconstruction probabilities of x by GMVAE_N and GMVAE_A are calculated, which reflect the models' reconstruction effects on the input data. Then, through logical fusion operation, the reconstruction probabilities of the input data from the two models are combined to determine whether an intrusion has occurred.

B. Stage1-Data Preprocessing

Raw network traffic data has many issues and must be preprocessed before being used as model input. The preprocessing steps in this paper are as follows:

- Data cleaning. The infinite values are replaced with -1, and the missing values are filled with the mean value.

- Data conversion. The traffic contains various string-type features (e.g., “state”, “proto”, “service”, etc.) that need to be encoded into numerical format. We use OneHotEncoder from scikit-learn for this conversion.
- Standardization. To address the diverse value ranges of continuous features and facilitate faster convergence of the gradient descent algorithm, we standardize the data using StandardScaler from scikit-learn.

C. Stage2-Data Reconstruction

We employ two GMVAEs to reconstruct traffic data. GMVAE_N is trained solely on normal traffic to construct a normal profile, while GMVAE_A is trained on known attack traffic to build an attack profile. The preprocessed traffic data x is simultaneously fed into the two GMVAEs for reconstruction. By analyzing the reconstruction results from these models, we can assess whether an attack has occurred. Specifically, if GMVAE_N has good reconstruction result while GMVAE_A has poor result, it indicates that the traffic is more likely to be normal; on the contrary, it may suggest that the traffic is more likely to be an attack. The detailed detection method will be explained in the following content.

D. Stage3-Detection

We use the reconstruction probability as the criterion, which indicates the likelihood of data being reconstructed by the model. Unlike reconstruction error, its normalized range between 0 and 1 allows for direct comparison of reconstruction quality across different data points or models, eliminating the impact of data magnitude. The reconstruction probability can be calculated as follows:

$$P_{reconstruction} = \frac{1}{L} \sum_{l=1}^L \sum_{i=0}^{k-1} p(y_i|x) P(x'_i|z_i^l) \quad (10)$$

where L is the number of Monte Carlo simulations, used to reduce the impact of sampling randomness; k is the number of Gaussian mixture components; $p(y_i|x)$ represents the probability that the input data belongs to the i -th Gaussian component; and $P(x'_i|z_i^l)$ is the probability of generating x'_i from the latent variable z_i^l in the l -th Monte Carlo sampling. We calculate the reconstruction probability p_n for the input data x using the

GMVAE_N model, and p_a for the same input data x using GMVAE_A model.

Algorithm 1 Logical Fusion

Require: reconstruction probabilities p_n and p_a , thresholds τ_n and τ_a
Ensure: ‘normal’ or ‘attack’

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1: if  $p_n \geq \tau_n$  and  $p_a \geq \tau_a$  then
2:    $result \leftarrow 'attack'$                                  $\triangleright$ Covert known attack
3: end if
4: if  $p_n \geq \tau_n$  and  $p_a < \tau_a$  then
5:    $result \leftarrow 'normal'$                                  $\triangleright$ Normal traffic
6: end if
7: if  $p_n < \tau_n$  and  $p_a \leq \tau_a$  then
8:    $result \leftarrow 'attack'$                                  $\triangleright$ New attack
9: end if
10: if  $p_n < \tau_n$  and  $p_a > \tau_a$  then
11:    $result \leftarrow 'attack'$                                  $\triangleright$ Explicit known attack
12: end if
13: return  $result$ 

```

Subsequently, we employ a logical fusion operation to obtain the detection results, as shown in Algorithm 1. p_n reflects the match degree of the tested traffic with normal traffic, while p_a reflects the match degree of the tested traffic with known attack traffic. By comparing the two reconstruction probabilities with their respective thresholds through logical fusion operations, four scenarios can be formed. i) When p_n exceeds the threshold τ_n and p_a exceeds the threshold τ_a , it means that the input data x matches both the characteristics of normal traffic and the characteristics of known attack traffic very well. In this case, we classify it as a **covert known attack**, i.e. a known attack type whose behaviour is highly similar to normal traffic and difficult to identify. ii) When p_n exceeds τ_n while p_a falls below τ_a , it indicates that the input data x is more consistent with the characteristics of normal traffic and has a lower match degree with known attack types. Therefore, we classify it as **normal traffic**. iii) When p_n is below τ_n and p_a is below τ_a , it means that the input data x does not match either the characteristics of normal traffic or the characteristics of known attack traffic. In this case, we classify it as a **new attack**. iv) When p_n is below τ_n while p_a is above τ_a , it means that the input data x does not match the characteristics of normal traffic but matches the characteristics of known attack traffic very well. In this case, we classify it as an **explicit known attack**, i.e. a known attack whose features are obvious and whose behaviour is significantly different from normal traffic.

V. PERFORMANCE EVALUATION

A. Experiment Setup

1) *Indicators:* We utilize several metrics to evaluate the performance of our proposed method, which are calculated using the True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

- **Accuracy.** The proportion of total samples that are predicted correctly.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \quad (11)$$

TABLE III
DATA TYPES AND SIZES OF CHD

Attack type	Number of records
Normal	988,872
DoS	587,521
Fuzzy	491,847
Spoofing gear	597,252
Spoofing RPM guage	654,897

- **Recall.** The proportion of actual positive samples that are correctly predicted as positive.

$$Recall = \frac{TP}{TP + FN}. \quad (12)$$

- **Precision.** The proportion of samples predicted as positive that are actually positive.

$$Precision = \frac{TP}{TP + FP}. \quad (13)$$

- **F1-Score.** The harmonic mean of precision and recall.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \quad (14)$$

- **False Positive Rate (FPR).** The proportion of real negative samples that are mistakenly predicted to be positive.

$$FPR = \frac{FP}{TN + FP}. \quad (15)$$

- **False Negative Rate (FNR).** The proportion of real positive samples that are mistakenly predicted to be negative.

$$FNR = \frac{FN}{FN + TP}. \quad (16)$$

2) *Description of Datasets:* The two datasets used to evaluate the performance of the proposed NIDS are: Car Hacking Dataset (CHD) [10] for evaluating detection of intrusions in in-vehicle communication and UNSW-NB15 Dataset (UBD) [11] for evaluating detection of intrusions generated from external networks. The description of the two datasets is as follows:

- **Car Hacking Dataset.** CHD provides authentic, real-world CAN traffic captured from a physical vehicle during actual message injection attacks, making it highly relevant for evaluating in-vehicle network intrusion detection. Table III shows the detailed information of CHD. The attack types include DoS, fuzzy, spoofing the drive gear, and spoofing the RPM gauge. The important features are extracted for use: CAN ID, DLC, DATA[0], DATA[1], DATA[2], DATA[3], DATA[4], DATA[5], DATA[6], DATA[7].
- **UNSW-NB15 Dataset.** UBD was developed to evaluate NIDS by constructing a synthetic laboratory environment using the IXIA PerfectStorm tool to generate a blend of normal and attack traffic. Given that real-world IoV attack data is extremely scarce due to privacy constraints and acquisition challenges, UBD's synthetic environment serves as a practical alternative. Although not originating from actual vehicles, its simulated traffic patterns closely resemble network interactions between vehicles

TABLE IV
DATA TYPES AND SIZES OF UBD

Attack type	Number of records
Normal	2,218,761
Exploits	44,525
Generic	215,481
DoS	16,353
Fuzzer	24,246
Reconnaissance	13,987

TABLE V
EXPERIMENTAL ENVIRONMENT CONFIGURATION

Software	Hardware
Ubuntu 20.04	Intel(R) Xeon(R) Gold 5218 CPU @ 2.30GHz
Python 3.6	NVIDIA GeForce RTX 4090
Pytorch 1.10.1+cu111	256GB RAM
Pycharm 2021.3	512GB ROM

and external communications. UBD includes network flow data of nine major families of attacks, which are also common in external network attacks on IoV. Table IV shows the details about the dataset used to evaluate the proposed NIDS. The dataset contains 49 features, such as source IP, destination IP, source port, destination port, duration, source bytes, and destination bytes.

3) *Implementation Details*: Table V displays the simulation environment. The experimental setup is shown in Table VI. We design nine scenarios across the two datasets to thoroughly assess our method's performance. Scenarios Sce@1 to Sce@5 are designed for the CHD, while Sce@5 to Sce@9 for UBD. Each scenario involves training two models: GMVAE_N, using only normal data, and GMVAE_A, using a subset of known attack data. During the testing phase, we evaluate the model's ability to detect known data types (normal data and attacks encountered during training) and new attacks (attacks that were not seen during training). New attacks are defined as attack types not included in the training dataset. These new attacks are designed to simulate real-world scenarios in which attackers may use tactics that have not been seen before. For example, in Table VI, for Sce@1, the attack types used during training are DoS, Fuzzy and Spoofing. Therefore, the spoofing RPM gauge is considered a new attack.

Model details are provided in Table VII. The models use fully connected layers, with structures specified in the 'structure' column. The selection of k is determined through a grid search method, with $k \in \{3, 4, 5, 6, 7\}$ and ELBO as the evaluation metric. For Sce@1 to Sce@4, k is set to 5 for GMVAE_N based on empirical findings, and to 3 for GMVAE_A, corresponding to the three known attack types during training. For Sce@5 to Sce@9, k remains at 5 for GMVAE_N, and is adjusted to 4 for GMVAE_A, corresponding to four known attack types in training. Additionally, we set the weighting factor α to 0.5, and use the Adam optimizer with a learning rate of 0.001. Each batch consists of 40 samples,

and the entire training process will go through 500 epochs. The reconstruction probability is calculated using Monte Carlo sampling, with the number of samples set to 10. Lastly, we set two detection thresholds: τ_n at 0.6 and τ_a at 0.9.

4) *Competitive Baseline Configuration*: We select four competitive baselines for fair and consistent performance comparison, all aligned with our research context and validated in state-of-the-art studies: OLGAM [18], BF+Bi-LSTM [25], PSSAE [29] and EBHM [19]. These methods represent the latest advances in intrusion detection research, and have garnered widespread attention owing to their superior performance in addressing complex network traffic patterns. The following is an overview of these approaches:

- OLGAM. Combines log-ratio oversampling, outlier detection, and metric learning for IoV intrusion detection, using LightGBM to achieve strong performance on datasets like UNSW-NB15.
- BF+Bi-LSTM. A multi-stage intrusion detection framework that uses state-based Bloom filters for initial anomaly screening and Bi-LSTM for temporal feature learning.
- PSSAE. A pseudo-siamese stacked autoencoder network that extracts deep semantic features via unsupervised pre-training and supervised fine-tuning for traffic classification.
- EBHM. A hybrid ensemble intrusion detection model addresses class imbalance by processing data through k-means clustering sampling, Z-score normalization, and SMOTE. It selects high-performance algorithms and fuses results via a stacking strategy.

All competitive baselines are re-implemented under a unified framework with identical scenario setup (Table VII), metrics (Section V-A1), and experimental environment (Table V), without specialized augmentation to ensure fairness.

B. Detection Performance

1) *Illustration of Detection Process*: Fig. 5 illustrates the detection process of EDGMVAE for various types of traffic in the Sce@1 scenario, displaying ten records per traffic type due to space limitations. Fig. 5a shows the detection of normal traffic, where p_n mostly exceeds the threshold τ_n , and p_a falls below τ_a , enabling accurate classification of normal traffic. According to the logical fusion algorithm, the sixth record, highlighted in green, indicates a false positive. Fig. 5b to Fig. 5d illustrate the detection process for known attacks, where it is observed that p_n is generally below τ_n while p_a exceeds τ_a . Notably, when p_n exceeds τ_n and p_a exceeds τ_a , we classify the traffic as an 'attack' that is similar to normal traffic. This ensures not only the detection of known attacks with obvious patterns but also those with covert patterns that resemble normal traffic. The records marked by green lines indicate false negatives. Finally, Fig. 5e illustrates the detection of new attacks, with both p_n and p_a not only below the thresholds. This aligns with our expectations and demonstrates that EDGMVAE effectively identifies most new attacks.

2) *Performance on CHD*: Table VIII summarizes the performance of the EDGMVAE on CHD. For normal traffic detection, EDGMVAE achieves an average accuracy rate of

TABLE VI
EXPERIMENTAL SETUP

Dataset	Scenario	Training data		Testing data	
		GMVAE_N	GMVAE_A	Known type	New attack
CHD	Sce@1	Normal	DoS, Fuzzy, Spoofing gear	Normal, DoS, Fuzzy, Spoofing gear	Spoofing RPM guage
	Sce@2	Normal	DoS, Fuzzy, Spoofing RPM guage	Normal, DoS, Fuzzy, Spoofing RPM guage	Spoofing gear
	Sce@3	Normal	DoS, Spoofing gear, Spoofing RPM guage	Normal, DoS, Spoofing gear, Spoofing RPM guage	Fuzzer
	Sce@4	Normal	Fuzzy, Spoofing gear, Spoofing RPM guage	Normal, Fuzzy, Spoofing gear, Spoofing RPM guage	DoS
UBD	Sce@5	Normal	Exploits, Generic, DoS, Fuzzer	Normal, Exploits, Generic, DoS, Fuzzer	Reconnaissance
	Sce@6	Normal	Exploits, Generic, DoS, Reconnaissance	Normal, Exploits, Generic, DoS, Reconnaissance	Fuzzer
	Sce@7	Normal	Exploits, Generic, Fuzzer, Reconnaissance	Normal, Exploits, Generic, Fuzzer, Reconnaissance	DoS
	Sce@8	Normal	Exploits, DoS, Fuzzer, Reconnaissance	Normal, Exploits, DoS, Fuzzer, Reconnaissance	Generic
	Sce@9	Normal	Generic, DoS, Fuzzer, Reconnaissance	Normal, Generic, DoS, Fuzzer, Reconnaissance	Exploits

TABLE VII
DETAILS OF PROPOSED MODEL

	Model	Structure	k	α	Learning rate	Batch	Epochs	L	Threshold
Sce@1 $\rightarrow$ Sce@4	GMVAE_N	Each layer of $NN(Q_y)$ has 512 nodes	5	0.5	0.001	40	500	10	$\tau_n = 0.6$ $\tau_a = 0.9$
	GMVAE_A	Each layer of $NN(Q_z)$ has 512 nodes	3						
Sce@5 $\rightarrow$ Sce@9	GMVAE_N	Each layer of $NN(P_z)$ has 64 nodes	5	0.5	0.001	40	500	10	$\tau_n = 0.6$ $\tau_a = 0.9$
	GMVAE_A	Each layer of $NN(P_x)$ has 512 nodes	4						

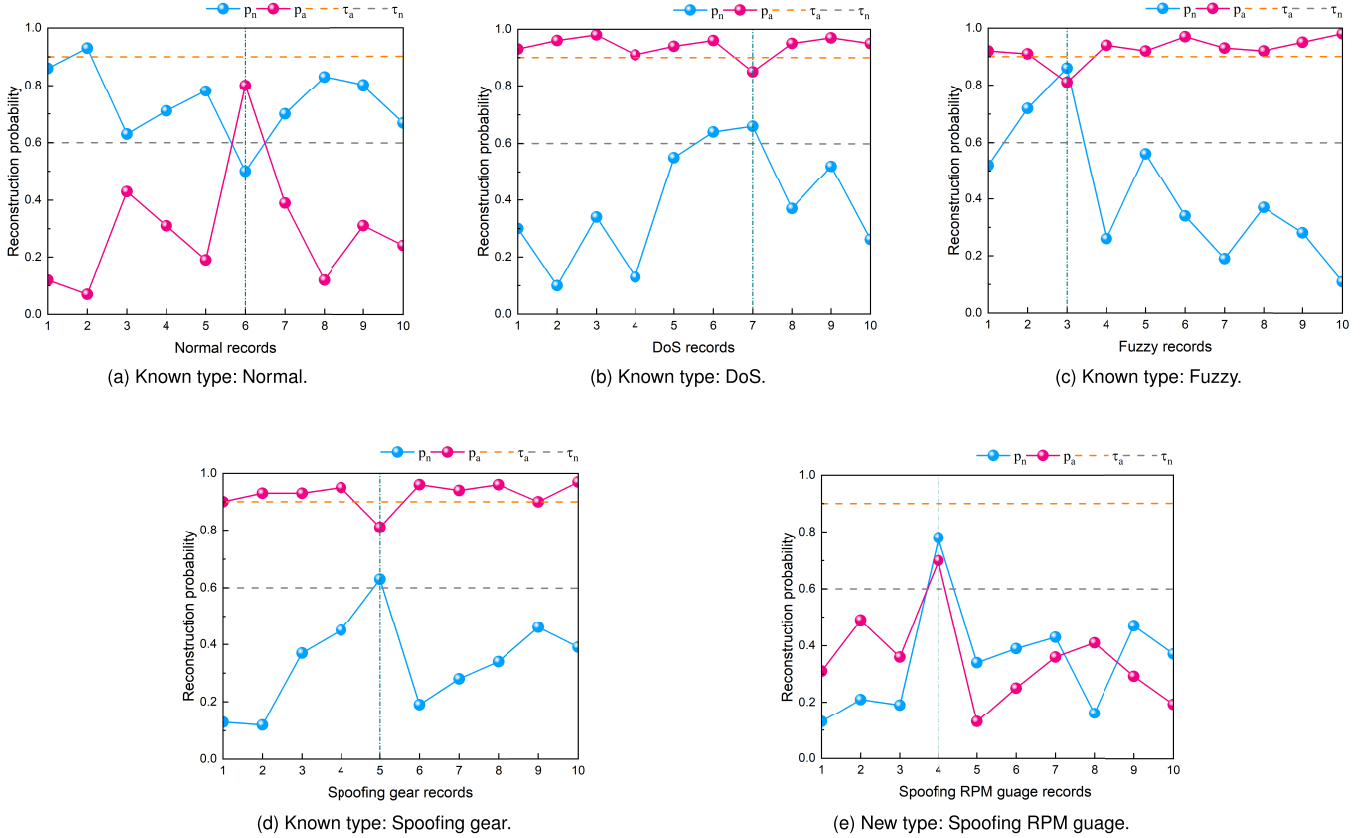


Fig. 5. Illustrating the detection process using Sce@1.

99.04% and an average false positive rate of merely 0.96%, indicating our method's high reliability in identifying normal traffic patterns. For known attacks, it attains an average

accuracy of 98.54%, recall of 98.52%, precision of 98.44%, and f1-score of 98.47%, with false negative rate of just 1.48%. These metrics highlight its strong performance in identifying

TABLE VIII
PERFORMANCE ON CHD

Scenario	Type		Accuracy(%)	Recall(%)	Precision(%)	F1-Score(%)	FNR(%)	FPR(%)
Sce@1	Known type	Normal	98.88	-	-	-	-	1.12
		DoS	99.15	98.26	98.47	98.36	1.74	-
		Fuzzy	97.67	98.71	98.24	98.47	1.29	-
		Spoofing gear	96.89	97.16	96.57	96.86	2.84	-
	New attack	Spoofing RPM guage	90.67	91.58	93.49	92.53	8.42	-
Sce@2	Known type	Normal	99.15	-	-	-	-	0.85
		DoS	98.35	98.38	97.99	98.18	1.62	-
		Fuzzy	98.79	98.91	99.12	99.01	1.09	-
		Spoofing RPM guage	99.27	98.66	99.03	98.84	1.34	-
	New attack	Spoofing gear	93.55	91.73	92.35	92.03	8.27	-
Sce@3	Known type	Normal	99.48	-	-	-	-	0.52
		DoS	99.19	99.06	98.93	98.99	0.94	-
		Spoofing gear	98.69	98.93	99.01	98.94	1.07	-
		Spoofing RPM guage	99.23	99.19	99.47	99.33	0.81	-
	New attack	Fuzzy	95.72	94.96	96.03	95.49	5.04	-
Sce@4	Known type	Normal	98.65	-	-	-	-	1.35
		Fuzzy	97.57	98.41	97.96	98.18	1.59	-
		Spoofing gear	98.67	97.39	97.91	97.65	2.61	-
		Spoofing RPM guage	98.99	99.17	98.53	98.85	0.83	-
	New attack	DoS	89.93	85.62	86.37	86.99	14.38	-

known attack patterns. For new attacks, EDGMVAE achieves an average accuracy of 92.47%, recall of 90.97%, precision of 92.06%, f1-score of 91.51%, and false negative rate of 9.03%. Despite the greater challenges posed by new attacks, these results highlight EDGMVAE's strong adaptability and potential for detecting new attack patterns.

3) *Performance on UBD*: Table IX presents the performance of EDGMVAE on UBD. For normal traffic detection, EDGMVAE achieves an impressive average accuracy of 98.91% and false positive rate of 1.09%. For known attacks, it attains an average accuracy of 98.59%, recall of 98.43%, precision of 98.40%, f1-score of 98.41%, and false negative rate of 1.57%. For new attacks, it achieves an average accuracy of 89.07%, recall of 89.85%, precision of 89.28%, f1-score of 89.56%, and false negative rate of 10.15%.

C. Ablation Study

We perform an ablation study in which we compare our method's performance with four ablative baselines:

- VAE_N: VAE trained exclusively on normal data, which classifies traffic as normal if the reconstruction probability exceeds the threshold τ_n , and as attack otherwise.
- VAE_A: VAE trained exclusively on known attack data, which identifies traffic as attack if the reconstruction probability surpasses the threshold τ_a , and as normal otherwise.
- GMVAE_N: GMVAE trained exclusively on normal data, which labels traffic as normal if the reconstruction

probability is above the threshold τ_n , and as attack if it is below.

- GMVAE_A: GMVAE trained exclusively on known attack data, which identifies traffic as attack if the reconstruction probability is greater than τ_a , and as normal otherwise.

We conduct a comprehensive comparison of our method with baseline methods across two datasets. Fig. 6 and Fig. 7 illustrate the detection performance of various models for normal traffic, known attack traffic, and new attack traffic on the CHD and UBD datasets, respectively.

1) Comparison on the Detection of Normal Traffic:

VAE_N and VAE_A exhibit relatively lower average accuracy and higher FPR in detecting normal traffic. This is because VAE models, using a single Gaussian distribution in the latent space, cannot accurately capture the complex underlying data features. In contrast, GMVAE_N and GMVAE_A exhibit a marked improvement in average accuracy. This enhancement can be credited to the use of Gaussian Mixture distributions as the latent space, which more effectively learns and represents the patterns in complex data. Overall, the results indicate that our EDGMVAE model, which integrates the strengths of GMVAE_N and GMVAE_A, achieves the highest accuracy and the lowest FPR.

2) Comparison on the Detection of Known Attack Traffic:

It is evident that VAE_A outperforms VAE_N, and GMVAE_A outperforms GMVAE_N. This can be attributed to the fact that VAE_A and GMVAE_A are specifically trained to learn the characteristics of known attacks, making them more sensitive

TABLE IX
PERFORMANCE ON UBD

Scenario	Type		Accuracy(%)	Recall(%)	Precision(%)	F1-Score(%)	FNR(%)	FPR(%)
Sce@5	Known type	Normal	99.28	-	-	-	-	0.72
		Exploits	98.91	96.76	97.87	97.31	3.24	-
		Generic	99.15	99.47	98.96	99.21	0.53	-
		DoS	99.64	99.17	98.61	98.89	0.83	-
		Fuzzer	99.04	98.73	99.01	98.87	1.27	-
	New attack	Reconnaissance	93.61	89.65	92.77	91.18	10.35	-
	Known type	Normal	97.93	-	-	-	-	2.07
		Exploits	98.26	95.62	97.38	96.49	4.38	-
		Generic	97.95	97.57	96.46	97.01	2.43	-
		DoS	98.94	98.94	97.71	98.32	1.06	-
		Reconnaissance	98.37	99.15	98.64	98.89	0.85	-
	New attack	Fuzzer	83.81	87.46	85.75	86.60	12.54	-
Sce@7	Known type	Normal	98.83	-	-	-	-	1.17
		Exploits	96.94	96.15	97.64	96.89	3.85	-
		Generic	97.85	98.14	98.06	98.10	1.86	-
		Fuzzer	98.81	98.46	97.97	98.21	1.54	-
		Reconnaissance	98.76	98.91	99.06	98.98	1.09	-
	New attack	DoS	88.45	89.73	90.14	89.93	11.55	-
	Known type	Normal	99.62	-	-	-	-	0.38
		Exploits	98.25	99.08	98.79	98.93	0.92	-
		DoS	99.67	99.63	99.57	99.60	0.37	-
		Fuzzer	98.15	98.99	99.02	99.00	1.01	-
		Reconnaissance	98.23	98.61	97.95	98.28	1.39	-
	New attack	Generic	94.17	92.76	91.23	91.99	7.24	-
Sce@9	Known type	Normal	98.89	-	-	-	-	1.11
		Generic	97.94	98.11	98.64	98.37	1.89	-
		DoS	99.37	99.26	98.71	98.98	0.74	-
		Fuzzer	98.94	99.24	98.76	99.00	0.76	-
		Reconnaissance	98.64	98.67	99.19	98.93	1.33	-
	New attack	Exploits	85.31	89.63	86.49	88.03	10.37	-

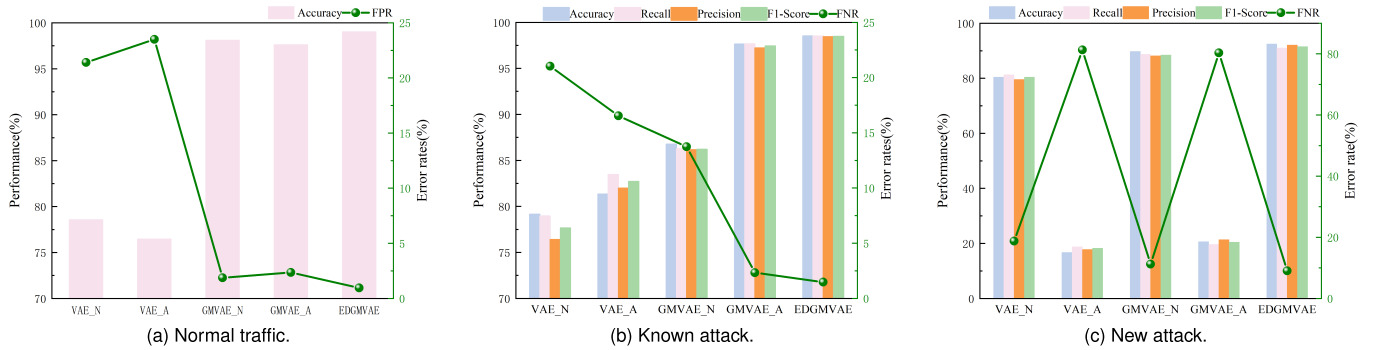


Fig. 6. Comparison of EDGMVAE with other baselines on the CHD.

to such patterns. GMVAE_N and GMVAE_A further demonstrate improvements over VAE_N and VAE_A, respectively, thereby highlighting the advantages of using Gaussian Mixture distribution as the latent space. Ultimately, EDGMVAE

achieves the best performance, confirming its reliability in detecting known attacks.

3) *Comparison on the Detection of New Attack Traffic:* VAE_A and GMVAE_A perform poorly in detecting new

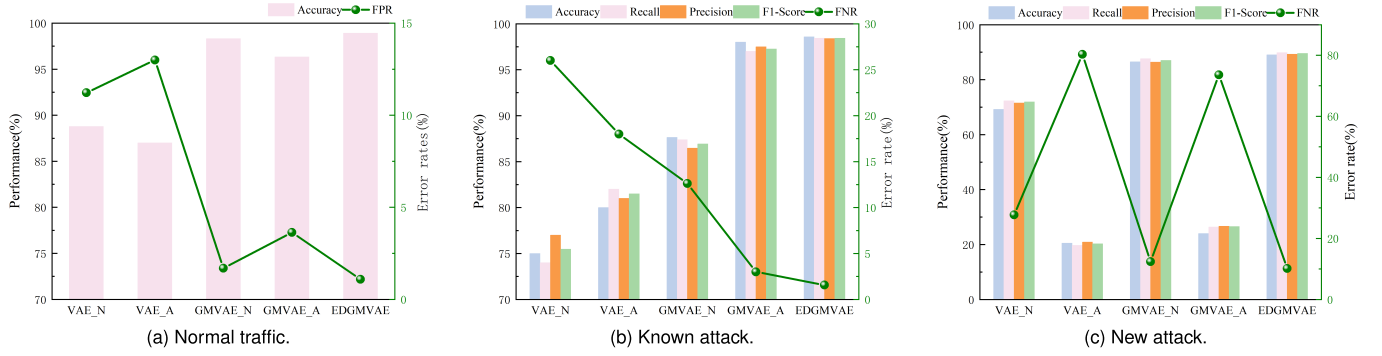


Fig. 7. Comparison of EDGMVAE with other baselines on the UBD.

TABLE X
COMPARISON WITH OTHER METHODS ON CHD

Method	Normal traffic		Known attack traffic					New attack traffic				
	Accuracy	FPR	Accuracy	Recall	Precision	F1-Score	FNR	Accuracy	Recall	Precision	F1-Score	FNR
OLGAM	97.64	2.36	96.13	96.24	96.67	96.45	3.76	79.64	77.36	78.87	78.11	22.64
BF+Bi-LSTM	94.82	5.18	96.57	95.43	96.74	96.08	4.57	85.81	85.67	84.29	84.97	14.33
PSSAE	96.76	3.24	97.62	96.99	97.07	97.03	3.01	82.37	83.14	82.96	83.05	16.86
EBHM	98.37	1.63	97.96	98.04	97.93	97.98	1.96	81.63	82.07	81.77	81.92	17.93
Proposed EDGMVAE	99.04	0.96	98.54	98.52	98.44	98.47	1.48	92.47	90.97	92.06	91.51	9.03

TABLE XI
COMPARISON WITH OTHER METHODS ON UBD

Method	Normal traffic		Known attack traffic					New attack traffic				
	Accuracy	FPR	Accuracy	Recall	Precision	F1-Score	FNR	Accuracy	Recall	Precision	F1-Score	FNR
OLGAM	98.24	1.86	97.03	97.27	98.24	97.75	2.73	67.16	65.48	65.72	65.60	34.52
BF+Bi-LSTM	95.31	4.69	98.54	97.73	98.61	98.16	2.27	83.94	85.57	83.62	84.09	15.43
PSSAE	97.29	2.71	98.62	98.34	97.99	98.16	1.66	75.61	74.68	76.61	75.63	25.32
EBHM	98.37	1.63	98.49	98.27	98.31	98.29	1.73	79.61	77.38	78.92	78.14	22.62
Proposed EDGMVAE	98.91	1.09	98.59	98.43	98.40	98.41	1.57	89.07	89.85	89.28	89.56	10.15

attacks, nearly failing in this regard. This is because they are trained only on known attack patterns and tend to misclassify unseen new attacks as normal traffic. In contrast, VAE_N and GMVAE_N show better performance in identifying new attacks, demonstrating a certain level of ability to recognize such patterns, although their FPR remains relatively high. EDGMVAE exhibits the most significant effectiveness in detecting new attacks. This confirms that our method maintains a high detection accuracy when faced with previously unseen attack patterns, showcasing its broad applicability and robustness in practical applications.

D. Computational Efficiency Analysis

To address the practical deployment feasibility of EDGMVAE in resource-constrained IoV environments, we analyze its computational complexity and time/space overhead, with comparisons to baseline models.

1) *Theoretical Complexity Analysis*: In the detection phase of EDGMVAE, the time complexity is determined by the key steps involved in computing reconstruction probabilities

using two parallel GMVAEs. Each GMVAE processes input traffic (with dimension d) through fully connected layers (with hidden layer size h) to extract latent features, involving matrix multiplications that contribute a complexity of $O(d \cdot h)$. For k Gaussian mixture components, the probability of the input belonging to each component is computed, adding $O(k \cdot h)$. To calculate the reconstruction probability $P_{\text{reconstruction}}$, L Monte Carlo samples are drawn, with each sample involving data reconstruction from latent variables and generation probability computation, contributing $O(L \cdot k \cdot d)$. Since the two GMVAE models run in parallel, the total time complexity for EDGMVAE inference remains $O(d \cdot h + k \cdot h + L \cdot k \cdot d)$. Space complexity is determined by the storage requirements for the parameters of the two GMVAEs, which is $O(2 \cdot (d \cdot h + h^2 + k \cdot h))$.

2) *Practical Overhead Measurements*: We measure the computational overhead using the experimental environment in Table V, with results in Table XII. We must explicitly acknowledge the gap between simulation and actual in-vehicle hardware; however, preliminary simulation-based results remain valuable for feasibility assessment because: (1) The relative overhead differences between methods

TABLE XII
PRACTICAL OVERHEAD MEASUREMENTS

	VAE	GMVAE	EDGMVAE
Training time(500 epochs)	5.1h	8.7h	17h
Detection latency(per traffic)	1.8ms	2.2ms	2.7ms
Peak memory usage	4.1GB	5.6GB	11.1GB

remain valid; (2) Modern AI-capable vehicular platforms (e.g., NVIDIA Drive AGX) provide enhanced AI inference optimization, which theoretically yields lower overhead than our simulation environment.

EDGMVAE takes 17 hours to train for 500 epochs, which is approximately twice as long as GMVAE. This training time is acceptable for IoV scenarios, given that intrusion detection models are usually trained offline prior to deployment and updated periodically rather than in real time. In terms of detection latency, EDGMVAE processes each traffic in 2.7ms, which is marginally longer than GMVAE (2.2ms) and VAE (1.8ms). EDGMVAE's peak memory usage is 11.1 GB, which is roughly twice that of GMVAE (5.6 GB), due to the storage of parameters for two GMVAEs. Thus, our method demonstrates feasibility in the IoV environment based on preliminary experimental evidence. Direct validation on in-vehicle hardware remains essential future work that we will actively pursue.

E. Comparison With Other Methods

Tables X and XI present the comparative results on the CHD and UBD, respectively. In the identification of normal traffic, compared with other methods, the false positive rate of EDGMVAE is 0.67% to 4.22% lower on the CHD and 0.54% to 3.6% lower on the UBD. In the detection of known attacks, EDGMVAE performs comparably with other methods and is generally superior to them. In the detection of new attacks, EDGMVAE shows significant advantages. Specifically, EDGMVAE achieves 6.66% to 12.83% higher accuracy and 5.3% to 13.61% lower false negative rate than other methods on the CHD, and achieves 5.13% to 21.91% higher accuracy and 5.28% to 24.37% lower false negative rate than other methods on the UBD.

VI. CONCLUSION AND FUTURE WORK

We propose the Enhanced Dual Gaussian Mixture Variational Autoencoder (EDGMVAE) to detect various network intrusions in the IoV, including new attacks. Our method consists of three main stages. In the initial stage, we preprocess the traffic data to prepare it for subsequent analysis. In the second stage, we reconstruct the traffic data using two distinct GMVAEs: GMVAE_N, trained on normal traffic, and GMVAE_A, trained on known attack traffic. We then obtain the reconstruction probabilities of the traffic data from both GMVAEs. By integrating these probabilities via a logical fusion procedure, we improve the model's capability to identify new attacks. We conduct tests on two benchmark datasets to evaluate the efficacy of our methodology: the

car hacking dataset for in-vehicle communication and the UNSWNB-15 dataset for external network communication. We carefully design various scenarios to thoroughly assess the model's ability. The evaluation results demonstrate that our method not only excels in detecting known attacks but also outperforms other state-of-the-art methods in detecting new attacks, highlighting its robustness and adaptability.

Despite the promising results achieved, we must acknowledge two main limitations of this study. First, experimental evaluation uses public datasets CHD and UBD with predefined, static attack patterns, which fail to fully capture real IoV adaptive attacks, potentially limiting generalization assessment to complex landscapes. Second, the logical fusion rules adopt fixed thresholds (τ_n, τ_a), which may compromise long-term detection stability under dynamic IoV traffic drifts (e.g., evolving normal behaviors or iterative attack strategies).

Future research will focus on the following directions: (1) Overcome predefined dataset limitations by extending evaluation to dynamic adaptive attacks, e.g., collecting real-road attack flow data with automakers or using adversarial generation to build strategy-iterating dynamic samples. (2) Design an online update mechanism to enable the model to adapt to the dynamic changes of the IoV environment. (3) Deploy EDGMVAE in large-scale IoV environments to further verify its efficiency and practical value. (4) Given that adversarial samples may bypass NIDS, it is necessary to conduct an in-depth evaluation of the robustness of the proposed NIDS when facing adversarial attacks.

REFERENCES

- [1] W. Zhuang, Q. Ye, F. Lyu, N. Cheng, and J. Ren, "SDN/NFV-empowered future IoV with enhanced communication, computing, and caching," *Proc. IEEE*, vol. 108, no. 2, pp. 274–291, Feb. 2020.
- [2] B. Ji et al., "Research on optimal intelligent routing algorithm for IoV with machine learning and smart contract," *Digit. Commun. Netw.*, vol. 9, no. 1, pp. 47–55, Feb. 2023.
- [3] H. Zhou, W. Xu, J. Chen, and W. Wang, "Evolutionary V2X technologies toward the Internet of Vehicles: Challenges and opportunities," *Proc. IEEE*, vol. 108, no. 2, pp. 308–323, Feb. 2020.
- [4] S. Rajapaksha, H. Kalutarage, M. O. Al-Kadri, A. Petrovski, G. Madzudzo, and M. Cheah, "AI-based intrusion detection systems for in-vehicle networks: A survey," *ACM Comput. Surv.*, vol. 55, no. 11, pp. 1–40, Nov. 2023.
- [5] K. Huang, R. Xian, M. Xian, H. Wang, and L. Ni, "A comprehensive intrusion detection method for the Internet of Vehicles based on federated learning architecture," *Comput. Secur.*, vol. 147, Dec. 2024, Art. no. 104067.
- [6] X. Li, Z. Hu, M. Xu, Y. Wang, and J. Ma, "Transfer learning based intrusion detection scheme for Internet of Vehicles," *Inf. Sci.*, vol. 547, pp. 119–135, Feb. 2021.
- [7] B. Lampe and W. Meng, "Intrusion detection in the automotive domain: A comprehensive review," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2356–2426, 4th Quart., 2023.
- [8] M. A. Alsoufi et al., "Anomaly-based intrusion detection systems in IoT using deep learning: A systematic literature review," *Appl. Sci.*, vol. 11, no. 18, p. 8383, Sep. 2021.
- [9] T. Saba, A. Rehman, T. Sadad, H. Kolivand, and S. A. Bahaj, "Anomaly-based intrusion detection system for IoT networks through deep learning model," *Comput. Electr. Eng.*, vol. 99, Apr. 2022, Art. no. 107810.
- [10] E. Seo, H. M. Song, and H. K. Kim, "GIDS: GAN based intrusion detection system for in-vehicle network," in *Proc. 16th Annu. Conf. Privacy, Secur. Trust (PST)*, Aug. 2018, pp. 1–6.
- [11] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–6.

- [12] M. D. Hossain, H. Inoue, H. Ochiai, D. Fall, and Y. Kadobayashi, "LSTM-based intrusion detection system for in-vehicle can bus communications," *IEEE Access*, vol. 8, pp. 185489–185502, 2020.
- [13] P. Wei, B. Wang, X. Dai, L. Li, and F. He, "A novel intrusion detection model for the CAN bus packet of in-vehicle network based on attention mechanism and autoencoder," *Digit. Commun. Netw.*, vol. 9, no. 1, pp. 14–21, Feb. 2023.
- [14] G. Xie, L. T. Yang, Y. Yang, H. Luo, R. Li, and M. Alazab, "Threat analysis for automotive CAN networks: A GAN model-based intrusion detection technique," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4467–4477, Jul. 2021.
- [15] F. Luo, J. Wang, Z. Li, and C. Luo, "An anomaly detection model for in-vehicle networks based on lightweight convolution with spectral residuals," *Comput. Secur.*, vol. 151, Apr. 2025, Art. no. 104304.
- [16] J. Shi, Z. Xie, L. Dong, X. Jiang, and X. Jin, "IDS-DEC: A novel intrusion detection for CAN bus traffic based on deep embedded clustering," *Veh. Commun.*, vol. 49, Oct. 2024, Art. no. 100830.
- [17] H. M. Song, J. Woo, and H. K. Kim, "In-vehicle network intrusion detection using deep convolutional neural network," *Veh. Commun.*, vol. 21, Jan. 2020, Art. no. 100198.
- [18] F. Jin, M. Chen, W. Zhang, Y. Yuan, and S. Wang, "Intrusion detection on Internet of Vehicles via combining log-ratio oversampling, outlier detection and metric learning," *Inf. Sci.*, vol. 579, pp. 814–831, Nov. 2021.
- [19] I. Ullah, I. Khalil, X. Bai, S. Garg, G. Kaddoum, and M. S. Hossain, "An ensemble-based hybrid model for the detection of attacks in the Internet of Vehicular Things," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 17914–17927, Oct. 2025.
- [20] H. M. Song and H. K. Kim, "Self-supervised anomaly detection for in-vehicle network using noised pseudo normal data," *IEEE Trans. Veh. Technol.*, vol. 70, no. 2, pp. 1098–1108, Feb. 2021.
- [21] J. Cao et al., "Anomaly detection for in-vehicle network using self-supervised learning with vehicle-cloud collaboration update," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 7454–7466, Jul. 2024.
- [22] L. Yang, A. Moubayed, and A. Shami, "MTH-IDS: A multitiered hybrid intrusion detection system for Internet of Vehicles," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 616–632, Jan. 2022.
- [23] S. Almutlaq, A. Derhab, M. M. Hassan, and K. Kaur, "Two-stage intrusion detection system in intelligent transportation systems using rule extraction methods from deep neural networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 15687–15701, Dec. 2023.
- [24] Y. Huang and M. Ma, "AAIL-IDS: An automatic incremental lifetime learning intrusion detection system for vehicular ad hoc networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 2, pp. 2669–2678, Feb. 2025.
- [25] I. A. Khan, N. Moustafa, D. Pi, W. Haider, B. Li, and A. Jolfaei, "An enhanced multi-stage deep learning framework for detecting malicious activities from autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 25469–25478, Dec. 2022.
- [26] S. Anbalagan, G. Raja, S. Gurumoorthy, R. D. Suresh, and K. Dev, "IIDS: Intelligent intrusion detection system for sustainable development in autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 15866–15875, Dec. 2023.
- [27] N. Nissar, N. Naja, and A. Jamali, "Securing VANETs: Multi-objective intrusion detection with variational autoencoders," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3867–3874, Feb. 2024.
- [28] M. Fu, P. Wang, M. Liu, Z. Zhang, and X. Zhou, "IoV-BERT-IDS: Hybrid network intrusion detection system in IoV using large language models," *IEEE Trans. Veh. Technol.*, vol. 74, no. 2, pp. 1909–1921, Feb. 2025.
- [29] S. Tu, M. Waqas, A. Badshah, M. Yin, and G. Abbas, "Network intrusion detection system (NIDS) based on pseudo-siamese stacked autoencoders in fog computing," *IEEE Trans. Services Comput.*, vol. 16, no. 6, pp. 4317–4327, Nov. 2023.



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